

Processing Big Data 2025/2026

Project

1) Project Scope

This course project adheres to the principles of active learning, where students integrate concepts learned during theoretical classes while also actively seeking complementary tools. During the project, students will have the chance to work with a multimodal dataset and are expected to come up with solutions to process and explore the data, trying to answer the question: “What do the data tell us?”.

2) Data Description

Debate Videos

Portugal went through the first round of presidential elections on January 18th 2026. During the two months before the election, candidates supported by the parties with political representation and an independent one participated in one-on-one debates, in a total of 28 debates across various Portuguese TV stations.

For each of the [debate videos](#), the following information was extracted:

Visual Features

All videos were sampled into frames, using a sample rate of 1 frame per second. For each of the frames, the following information was collected:

- **detected human poses** – a [pose detector](#) extracted
 - **location** of each detected person, expressed by a vector $[x1, y1, x2, y2]$, where $(x1,y1)$ is the position of the top left corner and $(x2,y2)$ corresponds to the bottom right one of a bounding box;
 - **class and confidence** of the detected person;
 - **poses** represented by the 2D coordinates of 17 keypoints for each detected human, as well as their probability of being visible, leading to a matrix of 17×3 for each pose;
- **detected faces** – a combination of a [face detector](#) with an [emotion recognition](#) model collected:
 - **location** of each detected face, expressed by a vector $[x1, y1, x2, y2]$ of bounding box coordinates;
 - **emotion** – a string with the expression/emotion predicted for each face;
 - **confidence scores** – the probability of the 8 possible emotion classes, associated to each face;
 - **facial landmarks** – a vector of 106-keypoints $[x1,y1,...,x106,y106]$ mapping to different locations in the face.

Audio and Speech Features

The audio track was split into short segments that contain a single speaker, using the [pyannote speaker-diarization](#) pipeline. The detected audio segments were processed to extract features that characterize the following aspects of speech:

- **pitch, harmonic to noise ratio (HNR), jitter, shimmer, formants, and formants-based features** - each represented by a scalar value, using the following [code](#);
- **speech-rate features** – number of pauses, speaking-rate, articulation-rate, speaking time, computed using the following [code](#);
- **speaker embedding vector** of dimension 512 extracted from the pyannote diarization deep learning model;
- **a transcript of the audio** performed by [Whisper](#). The transcripts correspond to the same audio segments as the audio features;
- **transcript embedding vector** of dimension 4096 extracted for each transcript using [EuroLLM](#), a multilingual large language model.

Text Features

- **supporting party** each candidate was mapped to his/her supporting party.
- **pool results** a timeline of pool results together with the agency responsible for conducting the interviews.

Newscast Videos

A set of prime-time newscast videos were also processed. These videos were collected from two different tv channels for the same days. Most of these videos are roughly aligned with a window of 7-10 days before the publication of a poll. The duration of the videos ranges from 35min to 1h and capture most of the news for a certain day.

For each of the [newscast videos](#), the following information was collected:

Visual/Text Features

All videos were sampled into frames, using a sample rate of 1 frame per second. For each of the frames, the following information was collected:

- **OCR** – a [optical character recognition](#) extracted
 - **locations** – vectors [x1, y1, x2, y2] with text bounding box positions;
 - **text** – the characters identified in the frames, mapped to words in Portuguese;
 - **confidence** – the confidence score of the recognized text.

Speech Features

As in the case of the debates, the audio track was split into short segments that contain a single speaker, using the [pyannote speaker-diarization](#) pipeline. The detected audio segments were processed to extract features

- a **transcript of the audio** performed by [Whisper](#);
- **transcript embedding vector** of dimension 4096 extracted for each transcript using [EuroLLM](#).

All information is stored in (pandas) dataframes, saved as a pickle file for each video: *_visual.pkl, *_audio.pkl, *_speech.pkl (for the segment transcripts), *_ocr.pkl*, and *_text.pkl for the remaining text features.

As an example, Figure 1 shows one frame overlayed with annotation of the detected objects (red), human poses (white), and faces (green). Figure 2 shows a transcript from the same video.

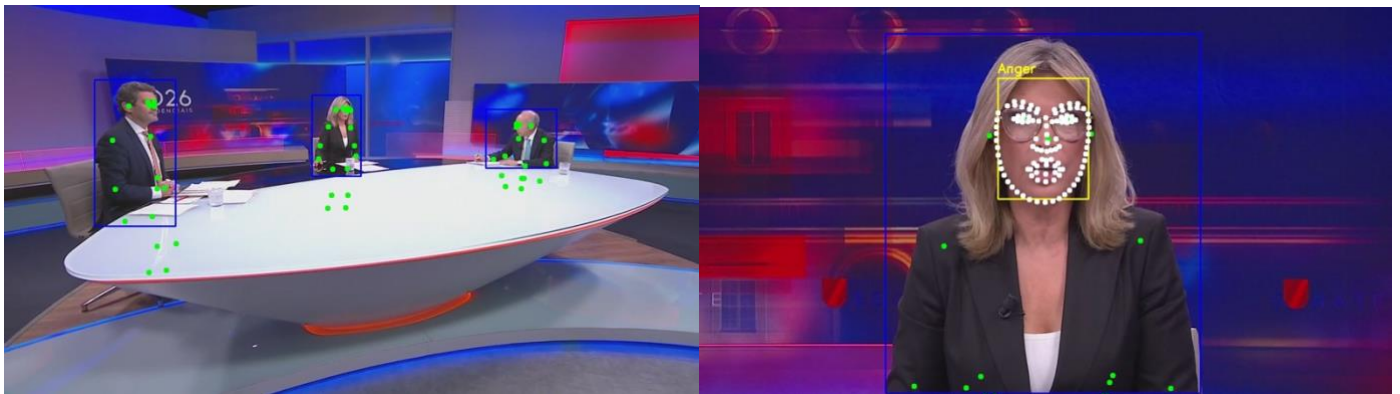


Figure 1: Example of some of the visual data extracted from the video frames.

E é o que tem feito mesmo no hospital, Marcelo Baldoso, a promulgou esta terça-feira, um diploma do governo. É, portanto, caso para dizer que está a trabalhar bem. Desmonte negro, toma a boa nota, da vitalidade do presidente, e não perde a oportunidade para recordar a qualidade do SNS. Também tivemos ocasião de verificar com o serviço nacional de saúde, fez a sua parte, fez com competência, fez com o espírito de serviço e demissão, que é o apanágio diário de muitos milhares de profissionais, que servem o nosso sistema de saúde, o serviço nacional de saúde também. As próximas visitas do Luís Montenegro e a Guia

Figure 2: Transcript by Whisper of a 46sec segment of a Newscast.

Não vai dizer agora que é que a vtro ter passado partidário. Como a pessoa que um dia se quer o Canadá, a gente vai publicar, tem que pensar muito bem e nunca militar nenhum partido, porque se não vai ser recusado no ser independente, os opuais não faz sentido absolutamente nenhum. E a despedência de razões, só é um ponto que também é importante. Independência da suas marcas, as marcas partidárias, também se vê nas pessoas que se que não me julgam, eu não tenho ninguém na minha candidatura

Figure 3: Transcript by whisper of a debate segment.

3) Objectives

The data described above represents a typical data science scenario, where we encounter large volumes of data originating from various types and sources. The goal of this project is to learn how to handle this type of real-world dataset and deal with the challenges of extracting meaningful information from it. The project is structured such that students are required to practice all steps of the data processing “pipeline”: Exploratory Data Analysis (EDA), Data Representation, Visualization, Modeling, Algorithm Design and Performance analysis. Furthermore, the project will show the reality of “big data processing”: lack of a clear problem structure, heterogeneous and multimodal data, large dimension, and missing or unreliable data.

To provide some guidance, the project is divided into three main tasks:

1. EDA & Visualization – familiarize yourselves with the data and perform exploratory data analysis and visualization by week 4 (to be evaluated during the midterm review)
2. Single Video Analysis – Choose **one debate** video and **one newscast** and process their data separately from the other videos.

Specific Newscast goals:

- a) Try to identify when the scene moves from the anchor to the different news.
- b) Try to identify different topics across the news, count the number of different news, and their length.

Specific Debate goals:

- a) Segment the debate into moments, for example explore the visual features and try to answer the following question “When does the camera capture a single debate participant, the other, and all?”.
- b) Identify the number of speakers in the video and try to map them across the various audio segments. You must also assign a label (“person 1”, “person 2”, etc.) to each detected voice segment and face in the video (whenever possible) and to assess how much (screen/voice) time was given to each participant.

- c) Try to find correlations between the different types of features, e.g., audio and the recognized emotions.

Both: Explore other interesting directions and try to answer the following question: is one modality enough to characterize the videos?

3. Multi-video analysis - Find additional interesting facts hidden in the data and across all videos (debates and newscast). What can you say about the candidates and their supportive parties from the collected data? Are there highly correlated features? Can you reduce the dimensionality of the data or use some of the features to predict others? Would you be able to predict the pool evolution? Can you relate topics discussed in the debates with those identified in the newscast? Is it possible to identify similar news across the two tv channels and do they appear in the same time window?

This task does not have a single clear solution or approach. Students are encouraged to explore and autonomously test multiple hypothesis/approaches/solutions and will be rewarded for their resourcefulness, creativity, and their outcomes or findings.

4) Evaluation

The Processing Big Data project should be done in groups of 4 students. It accounts for a total of 65% of the final grade, divided as follows: midterm review (15%), continuous evaluation during Thursday practical classes (10% - students are encouraged to attend the problem classes and make the most of the interaction with the professor), report and code evaluation (20%), presentation and discussion (20%).

Dates and deadlines

- The midterm review will take place during week 4 – this will consist of an 8min presentation + 7min discussion. (May 19-23 – The presentation slides must be submitted by Sunday, May 10th).
- The deadline for submission of the code is June 2nd (midnight).
- The deadline for submission of the 15min video report is June 6th (midnight).
- The presentation and discussion will take place during week 8 (June 8th-9th).